# An Empirical Study of an Edge-Aware Weisfeiler–Lehman Subtree Kernel

## Assignment 3 — Graph Machine Learning

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Dataset: MUTAG molecular graph classification benchmark

Report generated from measured cross-validation results

# Abstract

This project implements the Weisfeiler–Lehman (WL) subtree kernel from scratch and evaluates a bond-aware extension on the MUTAG molecular graph benchmark. The baseline refines node labels using neighbouring node labels; the modified method also incorporates chemical bond labels. A precomputed-kernel support vector machine is evaluated using repeated stratified 10-fold cross-validation. The reported results are generated by the included code, not manually estimated.

# 1. Introduction

Graph kernels compare graphs through structural patterns and let standard kernel classifiers operate on graph-structured data. Shervashidze et al. (2011) introduced an efficient WL graph-kernel framework based on iterative node-label refinement. This project tests whether including bond labels in the refinement step affects molecular classification.

# 2. Dataset

MUTAG contains 188 molecular graphs, 3371 nodes and 3721 edges in total. Nodes represent atoms and edges represent chemical bonds. The target is binary mutagenicity classification. The dataset is a standard benchmark, but its small size means the measured results have uncertainty.

# 3. Methods

Standard WL subtree kernel: initialise node labels from atom types, iteratively form signatures from each node's current label and the sorted multiset of neighbouring labels, compress identical signatures to shared identifiers, and count labels at each height. The graph kernel is the dot product of L2-normalised label-count feature vectors.

Edge-aware modification: each refinement signature includes the current node label and a sorted multiset of (bond label, neighbour label) pairs. A further ablation applies depth decay 0.7^h to reduce the weight of higher-order patterns. The implementation is from scratch and does not call a prebuilt WL-kernel library.

# 4. Experimental setup

All variants use an SVM with a precomputed kernel and C=1.0. Evaluation uses repeated stratified 10-fold cross-validation, 10 repeats, random seed 42. Metrics are averaged across test folds and include accuracy, balanced accuracy, macro-F1, ROC-AUC and runtime. The same protocol is used for every variant.

# 5. Results

| Method | Accuracy | Balanced acc. | Macro-F1 | ROC-AUC | Runtime (s) |
|---|---|---|---|---|---|
| WL subtree (h=0) | 0.665 ± 0.023 | 0.500 | 0.399 ± 0.008 | 0.693 ± 0.135 | 1.75 |
| WL subtree (h=1) | 0.750 ± 0.093 | 0.692 | 0.695 ± 0.111 | 0.861 ± 0.099 | 1.71 |
| WL subtree (h=2) | 0.785 ± 0.087 | 0.750 | 0.750 ± 0.101 | 0.879 ± 0.086 | 1.40 |
| WL subtree (h=3) | 0.792 ± 0.091 | 0.762 | 0.760 ± 0.104 | 0.883 ± 0.083 | 0.82 |
| WL subtree (h=4) | 0.787 ± 0.091 | 0.757 | 0.755 ± 0.104 | 0.886 ± 0.081 | 0.75 |
| Edge-aware WL (h=3) | 0.767 ± 0.087 | 0.693 | 0.699 ± 0.119 | 0.890 ± 0.082 | 0.78 |
| Edge-aware WL (h=3, decay=0.7) | 0.723 ± 0.065 | 0.612 | 0.601 ± 0.108 | 0.861 ± 0.099 | 0.75 |

At height 3, standard WL accuracy is 0.792 ± 0.091, macro-F1 is 0.760 ± 0.104, and ROC-AUC is 0.883 ± 0.083.

Edge-aware WL at height 3 obtains accuracy 0.767 ± 0.087, macro-F1 0.699 ± 0.119, and ROC-AUC 0.890 ± 0.082.

The difference in mean accuracy is -0.025; the difference in mean macro-F1 is -0.061. The best tested method by macro-F1 is WL subtree (h=3) (macro-F1 0.760).

# 6. Discussion

The edge-aware signature is more expressive because it distinguishes bond types when aggregating neighbourhood information. Greater expressiveness does not guarantee improved predictive performance: additional distinctions can help or add noise, especially with limited data. Interpret the observed changes as evidence on MUTAG rather than a universal result. Cross-validation reduces dependence on a single split but does not remove uncertainty.

# 7. Conclusion

The project implemented standard WL subtree features and an edge-aware variant, then compared them under a consistent SVM evaluation protocol. In this run, the edge-aware variant changed mean accuracy by -0.025 and macro-F1 by -0.061. The best configuration was WL subtree (h=3). These findings are limited to the MUTAG benchmark and the tested settings.

# References


[1] Shervashidze, N., Schweitzer, P., van Leeuwen, E. J., Mehlhorn, K., & Borgwardt, K. M. (2011). Weisfeiler-Lehman Graph Kernels. Journal of Machine Learning Research, 12, 2539–2561.

[2] Morris, C., Kriege, N. M., Kersting, K., & Mutzel, P. (2016). Faster Kernels for Graphs with Continuous Attributes via Hashing. IEEE ICDM 2016, 1095–1100.

[3] Togninalli, M., Ghisu, E., Llinares-López, F., Rieck, B., & Borgwardt, K. (2019). Wasserstein Weisfeiler-Lehman Graph Kernels. NeurIPS 2019.

[4] Debnath, A. K., et al. (1991). Structure-activity relationship of mutagenic aromatic and heteroaromatic nitro compounds. Journal of Medicinal Chemistry, 34(2), 786–797.